

37. (a) What is Tyndall effect ?

3

(b) What happens when sunlight is scattered from the particles of very large size ?

(c) 'Danger' signals are always red in colour. Why ?

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(b) What happens when sunlight is scattered from the particles of very large size ?

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(a)	The phenomenon of scattering of light by colloidal particles which makes the path of light visible is called Tyndall effect.	1
(b)	Scattered light may appear white. / Scattered light may appear of a colour of longer wavelength.	1
(c)	Red colour is least scattered by fog or smoke, so it can be seen in the same colour from a large distance.	1

36. What is scattering of light ? How is scattering of light different from reflection of light ? Explain.

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- The phenomenon of spreading light in different directions on interaction with particles of the medium.
- Light bounces back in a fixed direction after reflection while in scattering of light, it spreads in different directions. / Reflection of light is independent of the size of reflecting particles whereas the colour of the scattered light depends upon the size of scattering particles.

1

2

- (a) What happens when light is scattered from (i) very fine particles and (ii) very large size particles of medium ? Justify your answer in each case.

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2

(a) (i) Mainly blue coloured light is scattered because very fine particles scatter shorter wavelength.	1
(ii) Scattered light may appear white because very large size particles scatter light of all wavelengths.	1

- 35.** (a) Explain the Tyndall effect with the help of one suitable example.
- (b) What happens when a white light gets scattered from
(i) very fine particles (ii) larger size particles ?

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(a) • When a beam of light strikes fine particles like smoke, tiny water droplets, suspended particles of dust and molecules of air, the path of beam becomes visible. This phenomenon of scattering of light by colloidal particles is called Tyndall effect. • Examples: - ➤ When sunlight passes through a canopy of a dense forest tiny water droplets in the mist scatter light. ➤ When a fine beam of sunlight enters a smoke-filled room through a small hole, scattering of light makes the particles visible. (Any one, any other suitable example)	1 1
(b) (i) Blue light / light of shorter wavelength is visible. (ii) Red light/ Light of longer wavelength is visible.	$\frac{1}{2}$ $\frac{1}{2}$

- 25.** (a) When and where does a rainbow appear in the sky ? Draw a labelled ray diagram to show its formation. 2

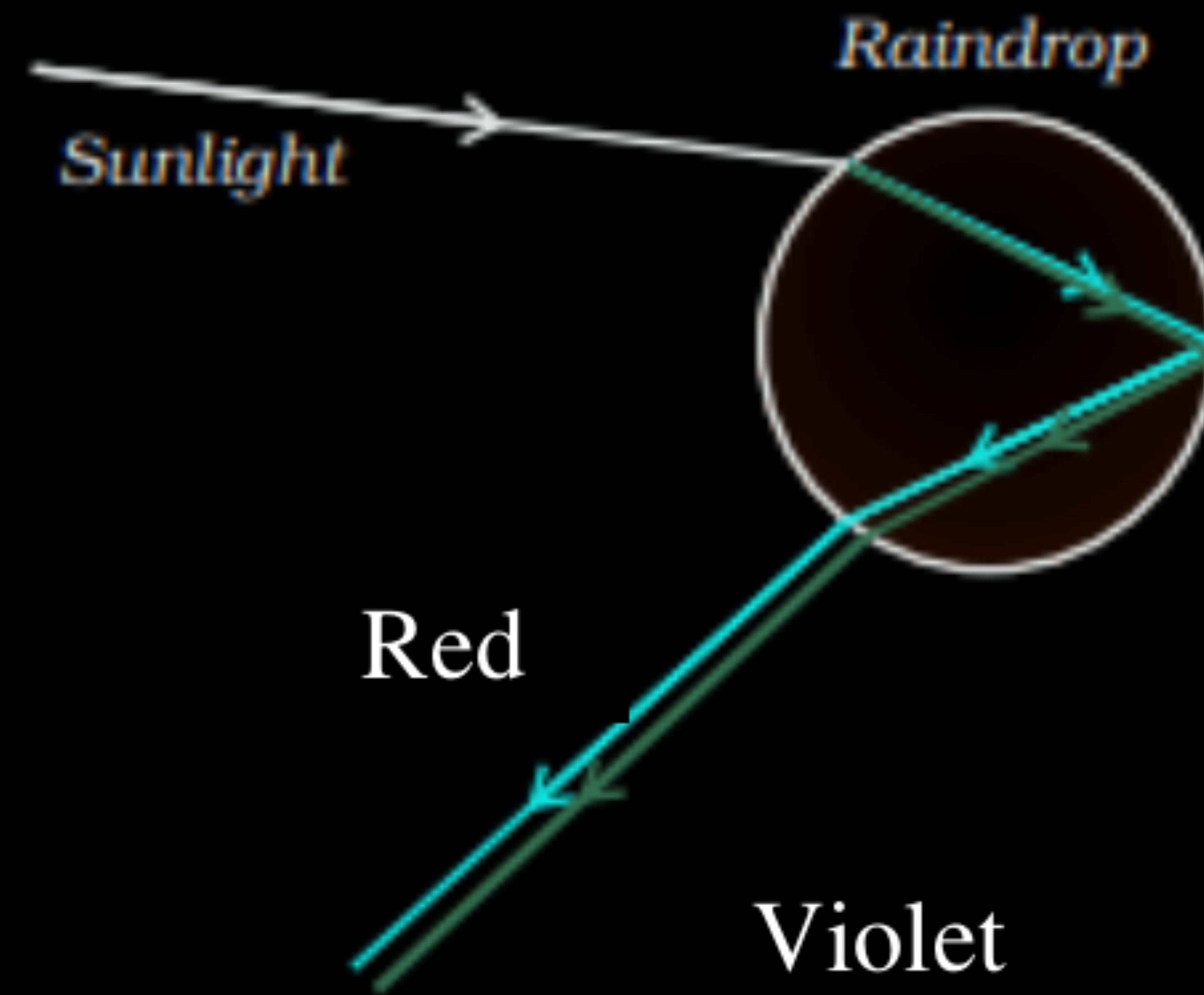
OR

- (b) What is scattering of light ? Why does the clear sky appear blue ? 2

- (a)
- It is formed after a rain shower.
 - It is always formed in a direction opposite to that of the Sun.

$\frac{1}{2}$

$\frac{1}{2}$



1

OR

- (b) (i) The phenomenon of the change in direction of propagation of light caused by large sized molecules/ caused by colloidal particles.

1

When sunlight passes through the atmosphere, fine particles in the air scatter blue light (shorter wavelengths) more than the red colour (longer wavelengths).

1

31. Give reasons for the following :

3

- (a) Danger signals installed at airports and at the top of tall buildings are of red colour.
- (b) The sky appears dark to the passengers flying at very high altitudes.
- (c) The path of a beam of light passing through a colloidal solution is visible.

31. Give reasons for the following :

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- (a) Danger signals installed at airports and at the top of tall buildings are of red colour.
- (b) The sky appears dark to the passengers flying at very high altitudes.
- (c) The path of a beam of light passing through a colloidal solution is visible.

(a) Red Coloured light is least scattered by fog or smoke and can be easily seen from a distance. / Red colour has high wavelength thus scattering is less.	1
(b) There is no scattering of light due to lack of atmosphere.	1
(c) Particles of colloid are big enough to scatter the beam of light.	1

(B) The colour of clear sky from the earth appears blue but from the space it appears black. Why ?

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2

(B) • Size of particles in the atmosphere is smaller than the wavelength of visible light, so they scatter light of shorter wavelengths i.e. blue. • In space, there is no scattering of light due to absence of particles. (no atmosphere)	1 1
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31. What is Tyndall effect ? Give two examples where this phenomenon is observed. Give reason to explain this phenomenon.

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• It is the phenomenon of scattering of light by the colloidal particles due to which the path of the light becomes visible. • Examples: 1. When sunlight passes through a canopy of dense forest. 2. A fine beam of sunlight enters a smoke/ dust filled dark room through a small hole. • Explanation: When a beam of light passes through a heterogenous mixture of minute particles (dust, smoke etc) it is reflected diffusely by the particles and gets scattered. This scattering of light makes the beam of light visible.	1 $\frac{1}{2}$ $\frac{1}{2}$ 1
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19. **Assertion (A)** : The colour of clear sky appears blue.

1

Reason (R) : Light of blue colour has longer wavelength as compared to the light of red colour so it is scattered more in the upper atmosphere.

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Reason (R) : Light of blue colour has longer wavelength as compared to the light of red colour so it is scattered more in the upper atmosphere.

(C) / (A) is true, but (R) is false

19. **Assertion (A)** : Red light signals are used to stop the vehicles on the road. **1**

Reason (R) : Red coloured light is scattered the most so as to be visible from a large distance.

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Reason (R) : Red coloured light is scattered the most so as to be visible from a large distance.

(C) / Assertion (A) is true, but Reason (R) is false

31. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke filled room through a small hole. Also state the dependence of colour of the light we receive on the size of the particle of the medium through which the beam of light passes.

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3

• Scattering of light / Tyndall effect	1
• When a beam of light strikes fine particles of smoke, it is reflected diffusely and the path of the light becomes visible.	1
• Very fine particles scatter mainly blue light/short wavelength colours while the particles of larger size scatter longer wavelength colours.	1

25. What would have been the colour of the sky, if the Earth had no atmosphere ? Give reason to justify your answer.

2

25. What would have been the colour of the sky, if the Earth had no atmosphere ? Give reason to justify your answer.

2

• Black colour	1
• As there would not be any particle to scatter light.	1

15. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is :

(A) Red

(B) Orange

(C) Blue

(D) Yellow

15. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is :

(A) Red

(B) Orange

(C) Blue

(D) Yellow

(C)/Blue

25. Name the phenomenon of light responsible for Tyndall effect. Write an event where this phenomenon can be observed.

2

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2

• Scattering of light.	1
• Example – When sunlight passes through a canopy of dense forest/ when a fine beam of sunlight enters a smoke filled room through a small hole. (or any other)	1

- (b) “The colour of the clear sky from the earth appears blue but the sky appears dark to the passengers flying at very high altitudes.”
Explain.

- (b) “The colour of the clear sky from the earth appears blue but the sky appears dark to the passengers flying at very high altitudes.” Explain.

2

(b) • Very fine particles of the atmosphere scatter short wavelength colour, mainly blue light, but scattering is not prominent at very high altitudes. So, sky appears black.	1+1
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23. (a) Give reasons :

(i) The sky appears dark to passengers flying at very high altitude.

(ii) 'Danger' signal lights are red in colour.

2

OR

(b) What is a rainbow ? "We see a rainbow in the sky only after the rainfall." Why ?

2

23. (a) Give reasons :

- (i) The sky appears dark to passengers flying at very high altitude.
- (ii) 'Danger' signal lights are red in colour.

2

OR

(b) What is a rainbow ? "We see a rainbow in the sky only after the rainfall." Why ?

2

(a) (i) Scattering of light is not prominent at such heights.

1

(ii) The red colour is least scattered by smoke or fog. / Red colour has longer wavelength.

1

OR

(b)

- Rainbow is a natural spectrum appearing in the sky after a rain shower.

1

After rain, small water droplets act as a tiny prism, when light enters, it gets refracted and dispersed.

1

19. **Assertion (A)** : The colour of the scattered light depends on the size of the scattering particles.

Reason (R) : Very fine particles scatter mainly red light while particles of larger size scatter light of shorter wavelengths.

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Reason (R) : Very fine particles scatter mainly red light while particles of larger size scatter light of shorter wavelengths.

(C)/ Assertion (A) is true, but Reason (R) is false.

12. Select the correct statement from the following :

- (A) The size of the molecules of air is larger than the wavelength of visible light.
- (B) The blue light has a wavelength about 1.8 times greater than that of red light.
- (C) When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red.
- (D) The light of red colour is scattered the most by fog or smoke.

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- (A) The size of the molecules of air is larger than the wavelength of visible light.
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- (D) The light of red colour is scattered the most by fog or smoke.

(C) / When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red.

12. When a beam of white light passes through a region of very fine dust particles, the colour of light that scatters the most in that region is :

(A) red

(B) orange

(C) blue

(D) yellow

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(C) /Blue

15. The phenomenon responsible for making the smoke particles visible when a beam of sunlight enters a smoke filled room through a narrow hole is :

- (A) scattering of light
- (B) dispersion of light
- (C) reflection of light
- (D) internal reflection of light

15. The phenomenon responsible for making the smoke particles visible when a beam of sunlight enters a smoke filled room through a narrow hole is :

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- (B) dispersion of light
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(A) scattering of light

Give reasons :

- (a) Red colour is selected for danger signals.
- (b) The sky appears dark in space.
- (c) The time difference between actual sunset and apparent sunset is about 2 minutes.

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- (a) Red colour is selected for danger signals.
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(a) The Red colour is least scattered by fog or smoke, hence visible from a long distance.	1
(b) Because in the absence of atmosphere there is no scattering of light.	1
(c) Because of atmospheric refraction, the sun appears above the horizon even after actual sunset.	1

21. Why is Tyndall effect shown by colloidal particles? State four instances of observing the Tyndall effect

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3

Because of scattering of light.

Instances:

- When a fine beam of light enters a smoke-filled dark room through a small hole.
- When sunlight passes through a canopy of dense forest in foggy/ misty conditions.
- Blue colour of sky.
- Red colour of the sun during sunrise or sunset.

(or any other)

1

$\frac{1}{2} \times 4$

22. Draw a labelled diagram to show (i) reddish appearance of the sun at the sunrise or the sunset and (ii) white appearance of the sun at noon when it is overhead.

22. Draw a labelled diagram to show (i) reddish appearance of the sun at the sunrise or the sunset and (ii) white appearance of the sun at noon when it is overhead.

3

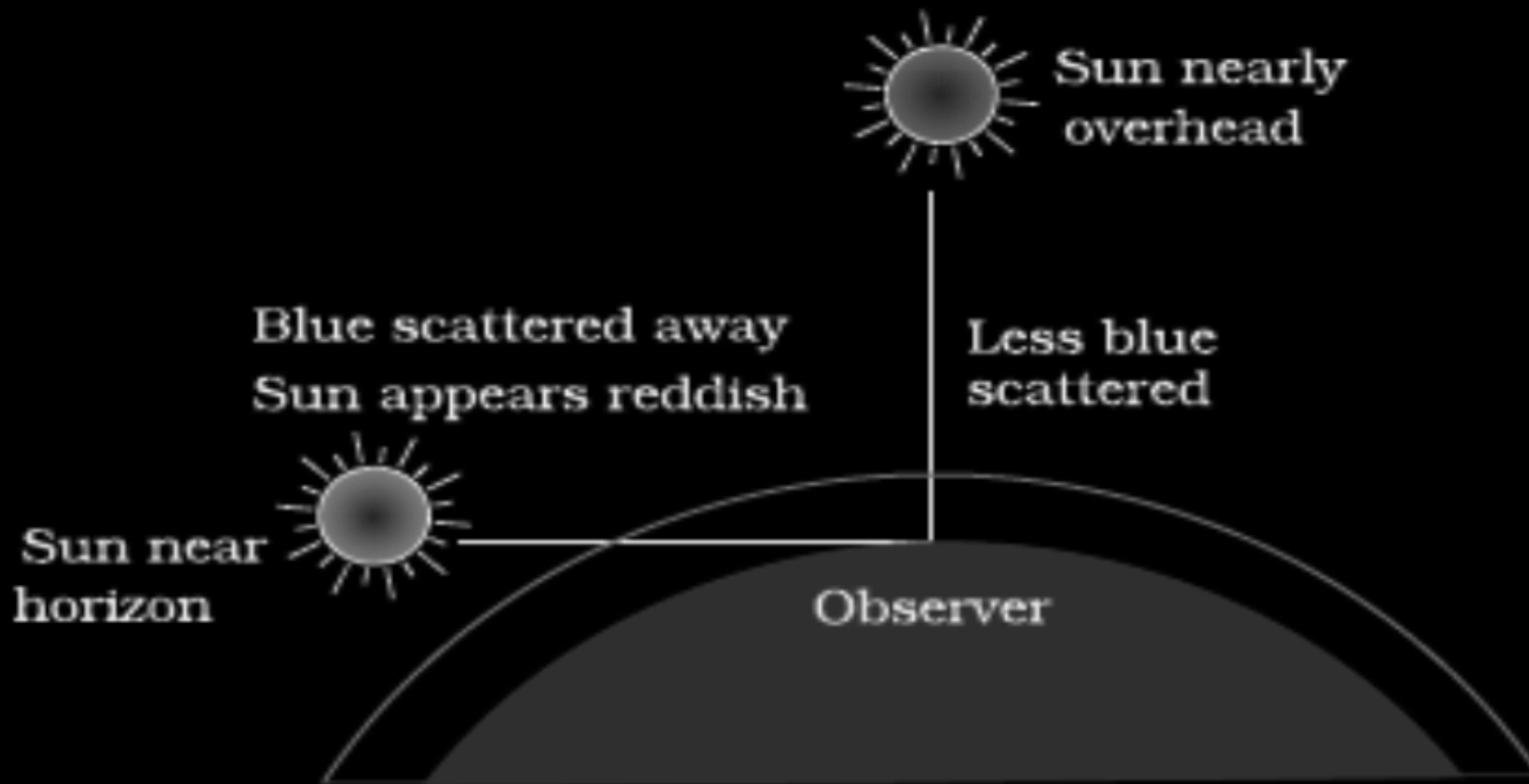


Diagram
Labelling

1
 $\frac{1}{2} \times 4$

- (a) What is scattering of light ? Explain how the colour of the scattered light depends on the size of the scattering particles.
- (b) Explain the reddish appearance of the Sun at sunrise or sunset. Why does it not appear red at noon ?

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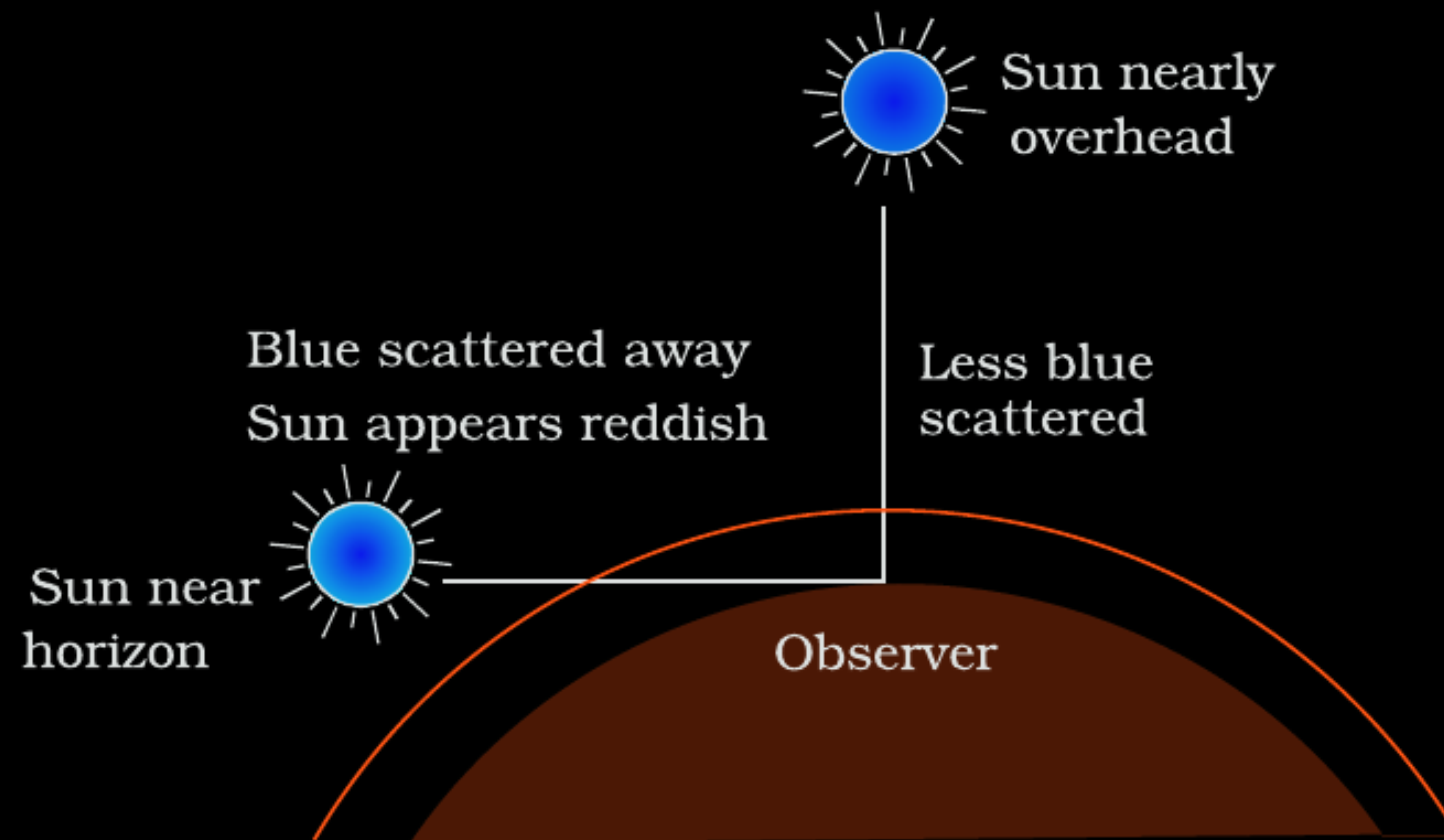


Figure 11.12
Reddening of the Sun at sunrise and sunset

(a) Scattering: Direction of ray of light changes when it collides with particles of comparable size.

- Fine particle scatter shorter wavelengths like blue light.
- Particles of larger size scatter longer wavelengths like red light.
- If particle size is large enough scattered light may appear white.

(b)

- During sunrise and sunset, sun is near horizon. Sunlight travels longer distance. Most of blue light scatters away by particles. Light of longer wavelength reaches our eye so sun appears reddish.
- At noon, sun is directly overhead so sunlight travels lesser distance. Less amount of blue light is scattered giving white appearance to sun

Or can be explained with the help of a diagram given below

1

$\frac{1}{2} \times 3$

$\frac{1}{2} \times 3$

1

What is atmospheric refraction ? List two phenomena which can be explained on the basis of atmospheric refraction.

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Refraction of light due to variation in optical density of earth.	
Two phenomenon:- (i) Twinkling of stars (ii) Advance sunrise & delayed sunset	$\frac{1}{2} \times 2$

Why is the colour of the clear sky blue ?

Why is the colour of the clear sky blue ?

When sunlight passes through the atmosphere, the fine particles in the air scatter the blue colour (short wavelength) more strongly than red. The scattered blue colour enters our eyes.	2
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What is atmospheric refraction ? Explain with the help of a labelled diagram that the position of a star as seen by us is not its true position.

What is atmospheric refraction ? Explain with the help of a labelled diagram that the position of a star as seen by us is not its true position.

- Bending of light due to the variation in optical density of the medium.
- The starlight, on entering into earth's atmosphere undergoes continuous refraction before it reaches the earth.
- The since the atmosphere bends starlight towards the normal, the apparent position to the star is slightly different from its actual position.

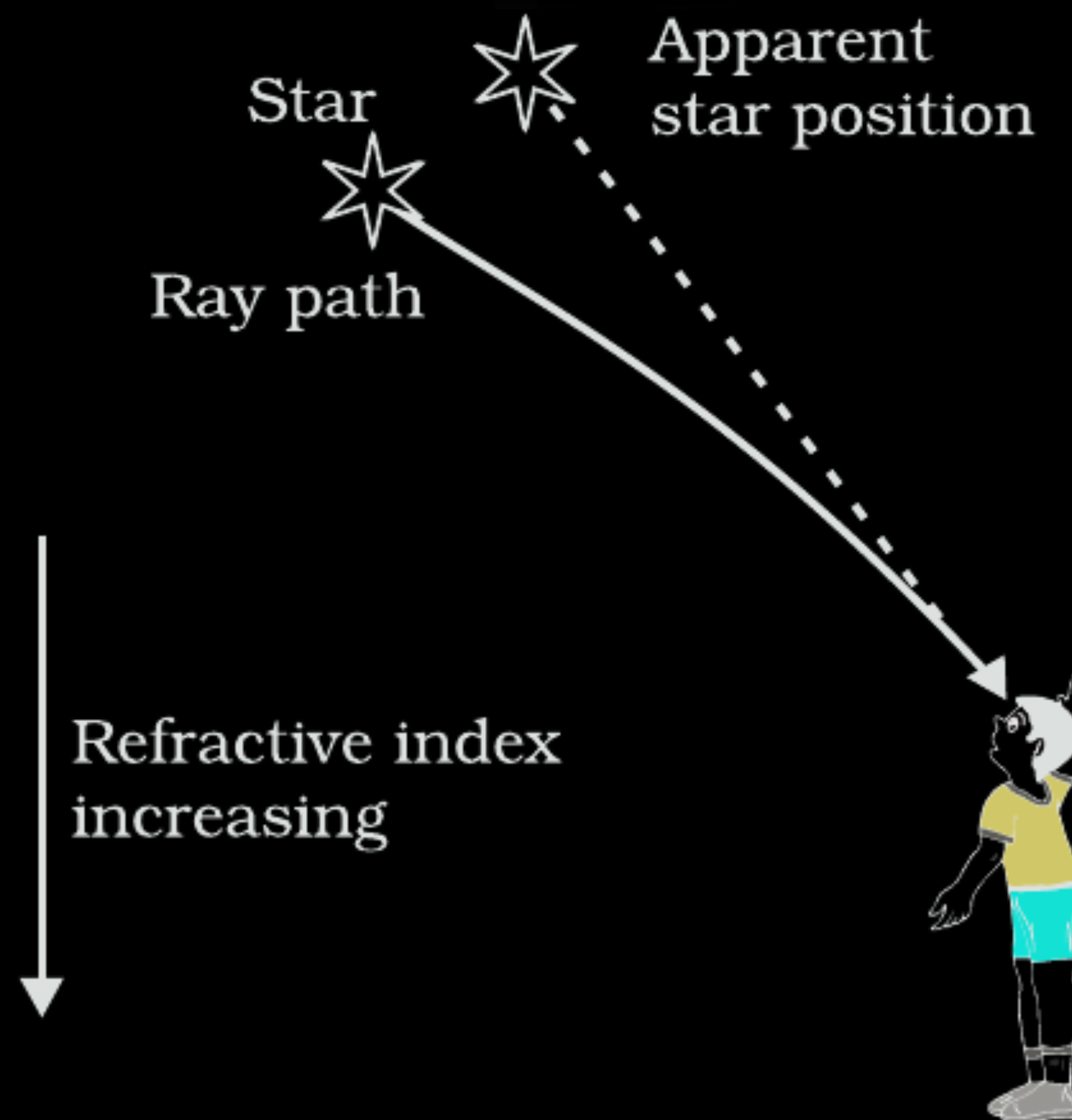


Diagram with Correct labeling

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$1\frac{1}{2}$

What is scattering of light ? Use this phenomenon to explain why (i) the Sun appears reddish at sun-rise, and (ii) the clear sky appears blue.

What is scattering of light ? Use this phenomenon to explain why (i) the Sun appears reddish at sun-rise, and (ii) the clear sky appears blue.

Scattering of light means to throw light in all possible direction when light intract with particles of medium.	1
(i)The Sun appears reddish at sun-rise : the Sun rays have to travel through a large atmospheric distance near the horizon .As the wave length of red light is maximum in the visible range , hence the scattering is least . The blue light and shorter wavelengths are scattered away by the particles .This gives rise to the reddish appearance of the sun./diagrammatic answers may be given fig:11.12	1
(ii)The sky appears blue: Blue colour has shorted wavelength than red. When sunlight passes through the atmosphere, the fine particles in the air scatter the blue light more strongly than red. Hence the clear sky spears blue.	1

What is atmospheric refraction ? Use this phenomenon to explain the following natural events.

- (a) Twinkling of stars
- (b) Advanced sun-rise and delayed sun-set.

Draw diagrams to illustrate your answers.

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(b) Advanced sun-rise and delayed sun-set.

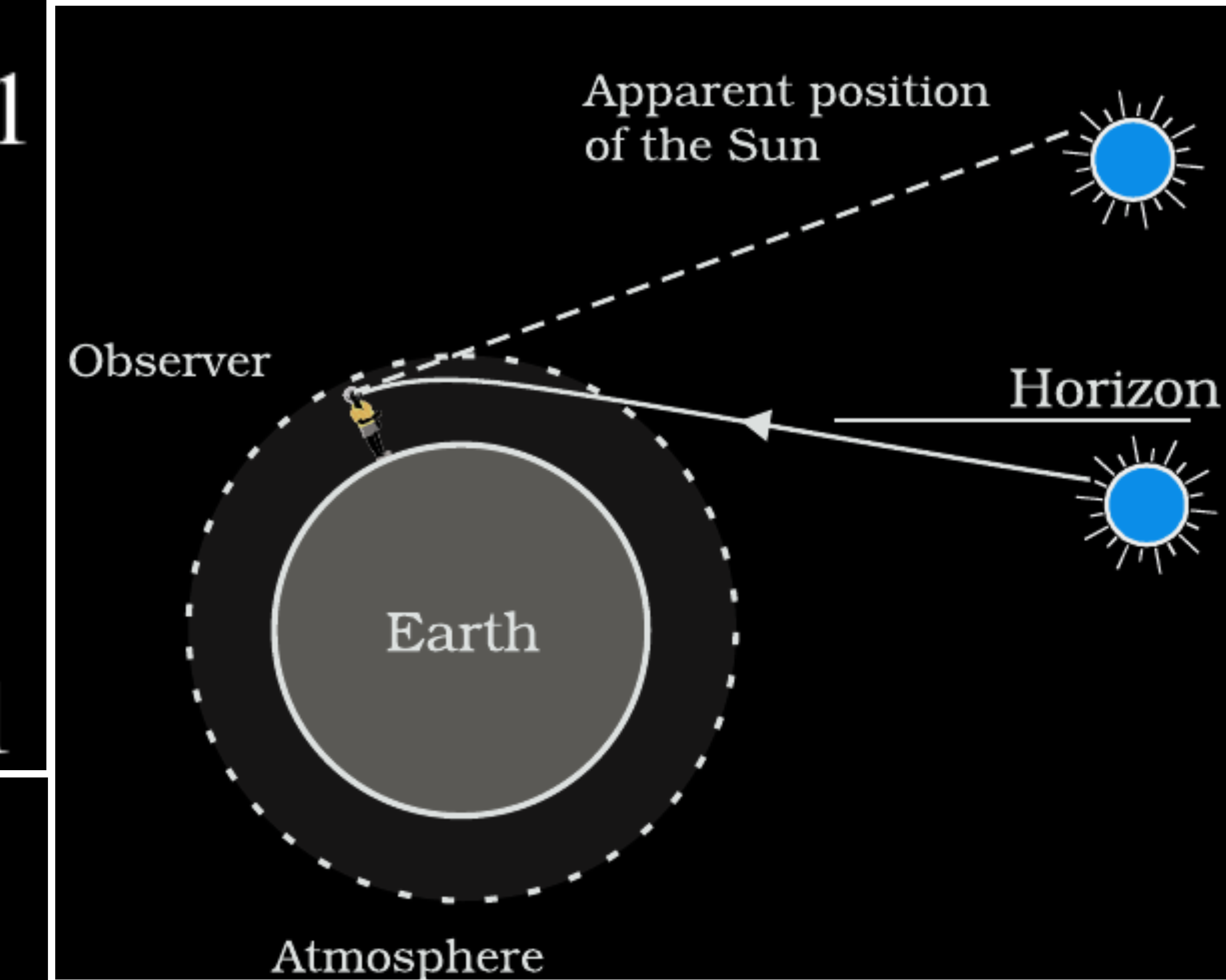
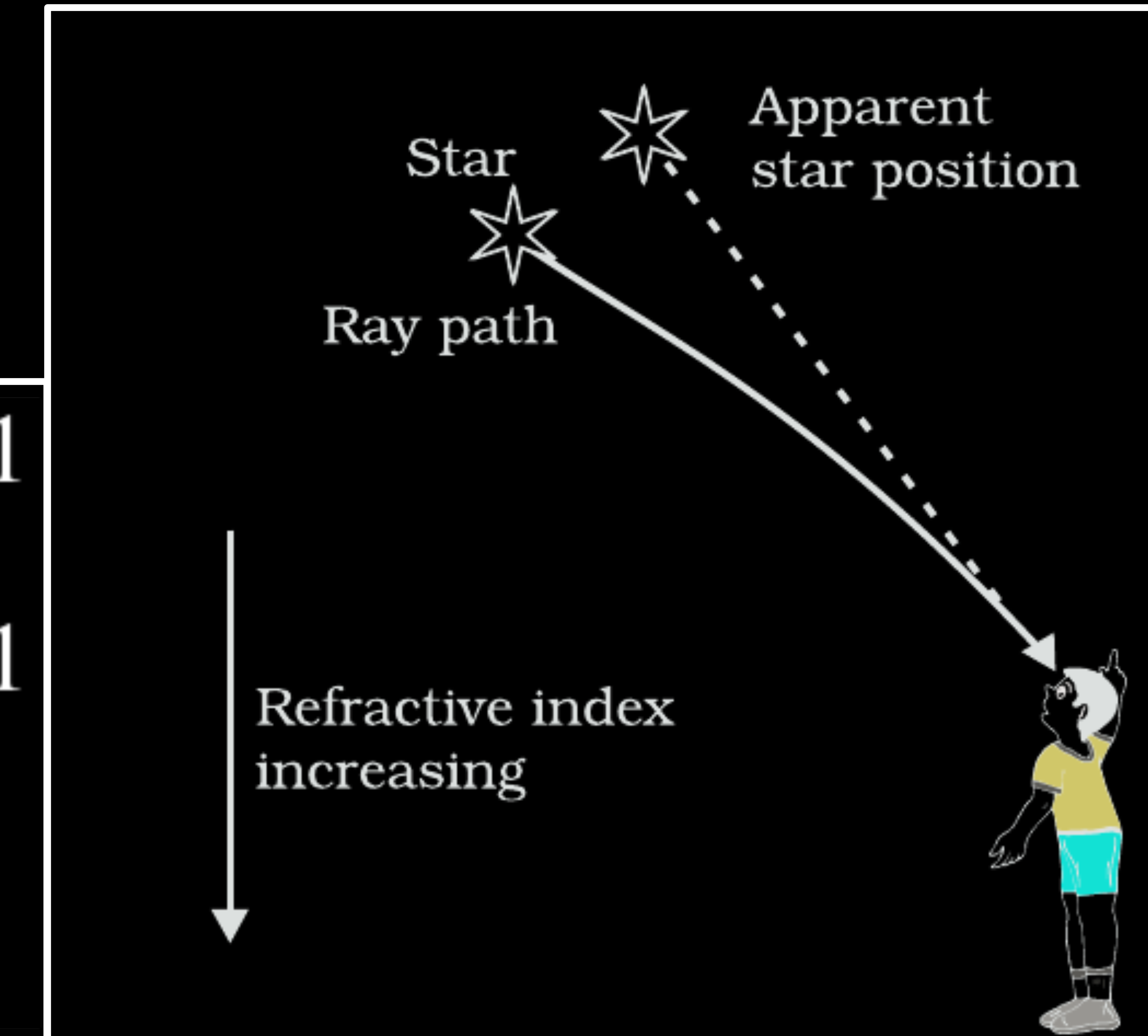
Draw diagrams to illustrate your answers.

Atmospheric refraction- refraction of light caused by the earth's atmosphere due to change in the refractive indices of different layers

Twinkling of stars- stars are distant point sized source of light. The path of the rays of light coming from the star goes on varying due to atmospheric refraction slightly. Thus apparent position of the stars fluctuates and the amount of star light entering the eye flickers giving the twinkling effect

Advanced sunrise – when the sun is slightly below the horizon, light rays coming from the sun travel from the rarer to denser layers of air. Because of atmospheric refraction of light, light appears to come from a higher position above the horizon. Thus sun appears earlier than actual sunrise.

Delayed sunset- Same reason as similar refraction occurs at the sunset.



What is meant by scattering of light? The sky appears blue and the sun appears reddish at sunrise and sunset. Explain these phenomena with reason.

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- | | |
|---|---|
| • Scattering of light – Phenomenon of spreading of light (diffused reflection) by minute particles in a medium. | 1 |
| • The sky appears blue because the blue colour of sunlight scatters much more strongly than the red colour by particles in atmosphere/ air due to its shorter wave length. | 1 |
| • At sun-rise and sun-set most of the blue light and shorter wavelengths are scattered away by the particles in the atmosphere as the light from the sun near the horizon passes through thick layers of air and larger distance. The light that reaches us is of longer wavelength (red colour) giving a reddish appearance. | 1 |

Why does the sun appear reddish early in the morning? Will this phenomenon be observed by an observer on the moon? Justify your answer with a reason.

Why does the sun appear reddish early in the morning? Will this phenomenon be observed by an observer on the moon? Justify your answer with a reason.

Early in the morning, the sun is near the horizon, sunlight reaches us after travelling a longer distance through thick layers of atmosphere. Thus most of the blue light and shorter wavelengths are scattered away by the particles in the atmosphere. The light that reaches us is of longer wavelengths giving a reddish appearance.

2

This phenomenon will not be observed by an observer on the moon.

$\frac{1}{2}$

Because of the absence of atmosphere on the moon.

$\frac{1}{2}$